

# ANSWER KEY

GATE EC — GATE EC 2026 Trend-Based Mock Paper

4

| Q.No. | Type | Answer | Marks |
|-------|------|--------|-------|
| 1     | 2MCQ | —      | 2     |
| 2     | 3MCQ | —      | 3     |
| 3     | 2MCQ | B      | 2     |
| 4     | 2MCQ | A      | 2     |
| 5     | 2MCQ | —      | 2     |
| 6     | 2NAT | —      | 2     |
| 7     | 3MCQ | —      | 3     |
| 8     | 2MCQ | —      | 2     |
| 9     | 2MSQ | —      | 2     |
| 10    | 2MSQ | —      | 2     |
| 11    | 2MCQ | B      | 2     |
| 12    | 2MCQ | —      | 2     |
| 13    | 2NAT | B      | 2     |
| 14    | 1NAT | B      | 1     |
| 15    | 1NAT | B      | 1     |
| 16    | 2MCQ | —      | 2     |
| 17    | 2NAT | —      | 2     |
| 18    | 1NAT | A      | 1     |
| 19    | 2MCQ | —      | 2     |
| 20    | 2MCQ | —      | 2     |
| 21    | 1MCQ | C      | 1     |
| 22    | 2NAT | —      | 2     |
| 23    | 1MCQ | B      | 1     |
| 24    | 1MCQ | B      | 1     |
| 25    | 1MCQ | B      | 1     |
| 26    | 1MCQ | B      | 1     |
| 27    | 1MCQ | B      | 1     |
| 28    | 1NAT | C      | 1     |
| 29    | 1MCQ | C      | 1     |
| 30    | 1NAT | B      | 1     |
| 31    | 1MCQ | A      | 1     |
| 32    | 1MCQ | A      | 1     |
| 33    | 1MCQ | B      | 1     |
| 34    | 1MCQ | A      | 1     |
| 35    | 1MCQ | A      | 1     |
| 36    | 1NAT | A      | 1     |

|    |      |   |   |
|----|------|---|---|
| 37 | 1MCQ | B | 1 |
| 38 | 1MCQ | A | 1 |
| 39 | 1NAT | B | 1 |
| 40 | 1MCQ | A | 1 |
| 41 | 1NAT | A | 1 |
| 42 | 1MCQ | A | 1 |
| 43 | 1MCQ | A | 1 |
| 44 | 1MCQ | C | 1 |
| 45 | 1MCQ | A | 1 |
| 46 | 1MCQ | B | 1 |
| 47 | 1MCQ | D | 1 |
| 48 | 2MCQ | — | 2 |
| 49 | 2MCQ | — | 2 |
| 50 | 2MCQ | — | 2 |
| 51 | 1MCQ | A | 1 |
| 52 | 2NAT | — | 2 |
| 53 | 2MCQ | — | 2 |
| 54 | 2MCQ | — | 2 |
| 55 | 1MCQ | B | 1 |
| 56 | 2MCQ | — | 2 |
| 57 | 2MCQ | — | 2 |
| 58 | 2NAT | — | 2 |
| 59 | 2NAT | — | 2 |
| 60 | 2MCQ | — | 2 |
| 61 | 2MCQ | — | 2 |
| 62 | 2MCQ | — | 2 |
| 63 | 2NAT | — | 2 |
| 64 | 1MCQ | B | 1 |
| 65 | 2MSQ | — | 2 |

**Q1.** General Aptitude | 2 marks | 2MCQ

Five years ago, the ratio of Aman's age to his father's age was 1:4, and five years from now, the ratio will be 2:5.

Wha...

**Answer: See explanation**

Source: Adapted from GATE 2024 EC Session 7 Q3

**Q2.** General Aptitude | 3 marks | 3MCQ

Which one of the following plots represents  $y = -||x|$ , where  $x$  is a non-zero real number? Note: The figures shown are rep...

**Answer: See explanation**

Source: Adapted from GATE 2025 EC Session 1 Q5

**Q3.** General Aptitude | 2 marks | 2MCQ

Sum of first 100 natural numbers:

**Answer: B**

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q10

**Q4.** General Aptitude | 2 marks | 2MCQ

Choose opposite of 'benevolent':

**Answer: A**

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q10

**Q5.** General Aptitude | 2 marks | 2MCQ

For a real number  $x > 1$ ,  $\frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3} + \frac{1}{x^4} = 1$  The value of  $x$  is

**Answer: See explanation**

Source: Adapted from GATE 2024 EC Session 7 Q4

**Q6.** General Aptitude | 2 marks | 2NAT

P, Q, R, S, and T have launched a new startup. Two of them are siblings. The office of the startup has just three rooms....

**Answer: See explanation**

Source: Adapted from GATE 2024 EC Session 7 Q2

**Q7.** General Aptitude | 3 marks | 3MCQ

The greatest prime factor of  $(3199 - 3196)$  is

**Answer: See explanation**

Source: Adapted from GATE 2024 EC Session 7 Q5

**Q8.** General Aptitude | 2 marks | 2MCQ

If 'f' denotes increasing order of intensity, then the meaning of the words [charm enamor bewitch] is analogous to fbores...

**Answer: See explanation**

Source: Adapted from GATE 2024 EC Session 7 Q1

**Q9.** General Aptitude | 2 marks | 2MSQ

The following figures show three curves generated using an iterative algorithm. The total length of the curve generated ...

**Answer: See explanation**

Source: Adapted from GATE 2025 EC Session 1 Q4

**Q10.** General Aptitude | 2 marks | 2MSQ

The 12 musical notes are given as  $\sharp, \sharp, \flat, \flat, \sharp, \flat, \flat, \sharp, \flat$ , and . Frequency of each note is 122 times the frequency of ...

**Answer: See explanation**

Source: Adapted from GATE 2025 EC Session 1 Q3

**Q11.** General Aptitude | 2 marks | 2MCQ

Choose synonym of 'trivial':

**Answer: B**

Correct Answer: B

Source: Adapted from GATE 2022 EC Session 1 Q10

**Q12.** General Aptitude | 2 marks | 2MCQ

Had I learnt acting as a child, I \_\_\_\_\_ a famous film star. Select the most appropriate option to complete the above ...

**Answer: See explanation**

Source: Adapted from GATE 2025 EC Session 1 Q2

**Q13.** Engineering Mathematics | 2 marks | 2NAT

Integral of  $1/x$  dx:

**Answer: B**

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q1

**Q14.** Engineering Mathematics | 1 marks | 1NAT

$i^4 =$ :

**Answer: B**

Correct Answer: B

Source: Adapted from GATE 2022 EC Session 1 Q1

**Q15.** Engineering Mathematics | 1 marks | 1NAT

$P(A \cap B) = ?$

**Answer: B**

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q1

**Q16.** Network, Signals & Systems | 2 marks | 2MCQ

In the given circuit, the current  $I_x$  (in mA) is \_\_\_\_\_. 1000  $\Omega$   $\rightarrow$  5 mA 1k 1k 1k  $\Omega$   $I_x$  2 mA

**Answer: See explanation**

Source: Adapted from GATE 2024 EC Session 1 Q31

**Q17.** Network, Signals & Systems | 2 marks | 2NAT

Two resistors are connected in a circuit loop of area 5 m<sup>2</sup>, as shown in the figure below. The circuit loop is placed on ...

**Answer: See explanation**

Source: Adapted from GATE 2025 EC Session 1 Q65

**Q18.** Network, Signals & Systems | 1 marks | 1NAT

Z-parameter conditions:

**Answer: A**

Correct Answer: A

Source: Adapted from GATE 2021 EC Session 1 Q8

**Q19.** Network, Signals & Systems | 2 marks | 2MCQ  
Let  $\mathbb{R}$  and  $\mathbb{C}$  denote the set of real numbers and the three dimensional vector space over it, respectively. The value of  $\dim(\mathbb{R} \otimes \mathbb{C})$  is ...

**Answer: See explanation**

Source: Adapted from GATE 2024 EC Session 1 Q30

**Q20.** Network, Signals & Systems | 2 marks | 2MCQ  
A 3-input majority logic gate has inputs  $x, y, z$  and  $w$ . The output of the gate is logic '1' if two or more of the inputs are ...

**Answer: See explanation**

Source: Adapted from GATE 2025 EC Session 1 Q23

**Q21.** Network, Signals & Systems | 1 marks | 1MCQ  
 $h$ -parameter represents:

**Answer: C**

Correct Answer: C

Source: Adapted from GATE 2023 EC Session 1 Q7

**Q22.** Network, Signals & Systems | 2 marks | 2NAT  
An electric field of  $0.01 \text{ V/m}$  is applied along the length of a copper wire of circular cross-section with diameter  $1 \text{ mm}$ . ...

**Answer: See explanation**

Source: Adapted from GATE 2025 EC Session 1 Q50

**Q23.** Electronic Devices | 1 marks | 1MCQ  
 $n$ -type majority carriers:

**Answer: B**

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q3

**Q24.** Electronic Devices | 1 marks | 1MCQ  
MOSFET is:

**Answer: B**

Correct Answer: B

Source: Adapted from GATE 2023 EC Session 1 Q8

**Q25.** Electronic Devices | 1 marks | 1MCQ  
SCR has:

**Answer: B**

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q9

**Q26.** Electronic Devices | 1 marks | 1MCQ  
Zener diode used for:

**Answer: B**

Correct Answer: B

Source: Adapted from GATE 2022 EC Session 1 Q9

**Q27.** Analog Circuits | 1 marks | 1MCQ  
Common emitter has:

**Answer: B**

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q4

**Q28.** Analog Circuits | 1 marks | 1NAT  
The voltage gain of a common-emitter amplifier is approximately

**Answer: C**

Correct Answer: C

Source: Adapted from GATE 2022 EC Session 1 Q3

**Q29.** Analog Circuits | 1 marks | 1MCQ  
Ideal op-amp open-loop gain:

**Answer: C**

Correct Answer: C

Source: Adapted from GATE 2023 EC Session 1 Q2

**Q30.** Analog Circuits | 1 marks | 1NAT  
FET is:

**Answer: B**

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q3

**Q31.** Analog Circuits | 1 marks | 1MCQ  
RC oscillator frequency:

**Answer: A**

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q9

**Q32.** Digital Circuits | 1 marks | 1MCQ  
4-bit binary counter counts:

**Answer: A**

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q4

**Q33.** Digital Circuits | 1 marks | 1MCQ  
D flip-flop follows input after:

**Answer: B**

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q5

**Q34.** Digital Circuits | 1 marks | 1MCQ  
ROM is:

**Answer: A**

Correct Answer: A

Source: Adapted from GATE 2021 EC Session 1 Q4

**Q35.** Digital Circuits | 1 marks | 1MCQ  
TTL stands for:

**Answer: A**

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q3

**Q36.** Control Systems | 1 marks | 1NAT  
*Routh-Hurwitz determines:*

**Answer: A**

Correct Answer: A

Source: Adapted from GATE 2024 EC Session 1 Q7

**Q37.** Control Systems | 1 marks | 1MCQ  
*Lead compensator:*

**Answer: B**

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q6

**Q38.** Control Systems | 1 marks | 1MCQ  
*Nyquist criterion uses:*

**Answer: A**

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q6

**Q39.** Control Systems | 1 marks | 1NAT  
*Transfer function is defined:*

**Answer: B**

Correct Answer: B

Source: Adapted from GATE 2023 EC Session 1 Q5

**Q40.** Communication Systems | 1 marks | 1MCQ  
*AM stands for:*

**Answer: A**

Correct Answer: A

Source: Adapted from GATE 2024 EC Session 1 Q6

**Q41.** Communication Systems | 1 marks | 1NAT  
*PCM stands for:*

**Answer: A**

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q4

**Q42.** Communication Systems | 1 marks | 1MCQ  
*FM has:*

**Answer: A**

Correct Answer: A

Source: Adapted from GATE 2021 EC Session 1 Q5

**Q43.** Communication Systems | 1 marks | 1MCQ  
*ASK modulation varies:*

**Answer: A**

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q5

**Q44.** Electromagnetics | 1 marks | 1MCQ  
Maxwell has how many equations:

**Answer: C**

Correct Answer: C

Source: Adapted from GATE 2024 EC Session 1 Q8

**Q45.** Electromagnetics | 1 marks | 1MCQ  
Characteristic impedance  $Z_0 =$ :

**Answer: A**

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q7

**Q46.** Electromagnetics | 1 marks | 1MCQ  
Waveguide supports:

**Answer: B**

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q7

**Q47.** Electromagnetics | 1 marks | 1MCQ  
Skin depth increases with:

**Answer: D**

Correct Answer: D

Source: Adapted from GATE 2023 EC Session 1 Q6

**Q48.** Network, Signals & Systems | 2 marks | 2MCQ  
In the circuit below, the opamp is ideal.  $R_1 = C$ ,  $R_2 = R + R$ ,  $C$ . If the circuit is to show sustained oscillations, the respect...

**Answer: See explanation**

Source: Adapted from GATE 2024 EC Session 1 Q50

**Q49.** Network, Signals & Systems | 2 marks | 2MCQ  
Which of the following statements is/are TRUE with respect to ideal MOSFET-based DC-coupled single-stage amplifiers hav...

**Answer: See explanation**

Source: Adapted from GATE 2025 EC Session 1 Q30

**Q50.** Network, Signals & Systems | 2 marks | 2MCQ  
For a causal discrete-time LTI system with transfer function  $H(z) = (2z^2 + 3z + 13) / (z^2 - 13z + 13)$  - which of the following statements ...

**Answer: See explanation**

Source: Adapted from GATE 2024 EC Session 1 Q24

**Q51.** Network, Signals & Systems | 1 marks | 1MCQ  
FS exists for:

**Answer: A**

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q2

**Q52.** Network, Signals & Systems | 2 marks | 2NAT  
The free electron concentration profile  $n(x)$  in a doped semiconductor at equilibrium is shown in the figure, where the  $p...$

**Answer: See explanation**

Source: Adapted from GATE 2024 EC Session 1 Q26

**Q53.** Network, Signals & Systems | 2 marks | 2MCQ  
Consider a system  $S$  represented in state space as  $\dot{x} = 0 - 2x + 1r$ ,  $y = 2 - 5x$ .  $\det 1 - 3 \ 0$  Which of the state space rep...

**Answer: See explanation**

Source: Adapted from GATE 2024 EC Session 1 Q53

**Q54.** Network, Signals & Systems | 2 marks | 2MCQ  
Two fair dice (with faces labeled 1, 2, 3, 4, 5, and 6) are rolled. Let the random variable denote the sum of the outcom...

**Answer: See explanation**

Source: Adapted from GATE 2025 EC Session 1 Q58

**Q55.** Electronic Devices | 1 marks | 1MCQ  
 $n$ -type majority carriers:

**Answer: B**

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q3

**Q56.** Network, Signals & Systems | 2 marks | 2MCQ  
The general form of the complementary function of a differential equation is given by  $y(t) = (At + B)e^{-2t}$ , where  $A$  an...

**Answer: See explanation**

Source: Adapted from GATE 2024 EC Session 1 Q11

**Q57.** Network, Signals & Systems | 2 marks | 2MCQ  
An amplitude modulator has output (in Volts)  $s(t) = A\cos(400t) + B\cos(360t) + B\cos(440t)$ . The carrier power normalize...

**Answer: See explanation**

Source: Adapted from GATE 2024 EC Session 1 Q28

**Q58.** Network, Signals & Systems | 2 marks | 2NAT  
In the circuit given below, the switch  $S$  was kept open for a sufficiently long time and is closed at time  $t = 0$ . The time ...

**Answer: See explanation**

Source: Adapted from GATE 2024 EC Session 1 Q32

**Q59.** Network, Signals & Systems | 2 marks | 2NAT  
For a causal discrete-time LTI system with transfer function  $H(z) = (2z^2 + 3z + 13)/(z^2 - 13)$  - which of the following statements ...

**Answer: See explanation**

Source: Adapted from GATE 2024 EC Session 1 Q24

**Q60.** Network, Signals & Systems | 2 marks | 2MCQ  
Let  $X(t) = A\cos(2\pi f_0 t + \phi)$  be a random process, where amplitude  $A$  and phase  $\phi$  are independent of each other, and are unifo...

**Answer: See explanation**

Source: Adapted from GATE 2024 EC Session 1 Q57

**Q61.** Network, Signals & Systems | 2 marks | 2MCQ  
As shown in the circuit, the initial voltage across the capacitor is 10 V, with the switch being open. The switch is the...

**Answer: See explanation**

Source: Adapted from GATE 2024 EC Session 1 Q35

**Q62.** Network, Signals & Systems | 2 marks | 2MCQ

The opamps in the circuit shown are ideal, but have saturation voltages of  $\pm 10$  V. Assume that the initial inductor current...

**Answer: See explanation**

Source: Adapted from GATE 2024 EC Session 1 Q49

**Q63.** Network, Signals & Systems | 2 marks | 2NAT

and are Bernoulli random variables taking values in  $\{0,1\}$ . The joint probability mass function of the random variables  $I_1, I_2$  is...

**Answer: See explanation**

Source: Adapted from GATE 2025 EC Session 1 Q60

**Q64.** Control Systems | 1 marks | 1MCQ

Lead compensator:

**Answer: B**

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q6

**Q65.** Network, Signals & Systems | 2 marks | 2MSQ

Consider a non-negative function  $f(t)$  which is continuous and bounded over the interval  $[2, 8]$ . Let  $a$  and  $b$  denote, respectively...

**Answer: See explanation**

Source: Adapted from GATE 2025 EC Session 1 Q51